

Analysis of urban impacts on aquatic habitats in the central Amazon basin: Adult odonates as bioindicators of environmental quality



Cláudio da Silva Monteiro Júnior^{a,b,*}, Leandro Juen^b, Neusa Hamada^a

^a Coordenação de Biodiversidade, Instituto Nacional de Pesquisas da Amazônia, Av. André Araújo, 2936, Petrópolis, CEP 69067-375 Manaus, Amazonas, Brazil

^b Laboratório de Ecologia e Conservação, Instituto de Ciências Biológicas, Universidade Federal do Pará, Rua Augusto Correia, N° 1 Bairro Guamá, CEP 66.075-110 Belém, Pará, Brazil

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ABSTRACT

Thirty streams were surveyed in urban and natural settings in the municipality of Manaus in the central Amazon basin (Brazil) with the objective of identifying the species of adult odonates that can be used as bioindicators of environmental quality. The data collected were used to test the hypothesis that species in the suborder Zygoptera are indicators of better-preserved environments due to their smaller body sizes and reduced tolerances to habitat modification, whereas species in the suborder Anisoptera were presumed to be indicators of impacted habitats with no vegetation. The habitats were classified as preserved, intermediate, and degraded, based on their environmental characteristics. A total of 908 specimens were collected, representing 60 species. The results of the indicator value (IndVal) identified 13 species as indicators of environmental quality, of which nine were typical of preserved habitats, two of intermediate habitats, and four of modified habitats (intermediate or degraded). Odonate species richness declined with increasing urbanization, a pattern also presented by the zygopterans, although anisopteran species richness was higher in intermediate habitats. Zygopteran species showed high fidelity/specificity for preserved habitats, although a small number of the species of this suborder showed a similar relationship with intermediate or degraded habitats, whereas anisopterans were associated only with disturbed habitats (intermediate and degraded). Overall, the results indicate that the diagnosis of the adult odonate community can provide a rapid and effective tool for evaluation of environmental quality. As many species are stenotopic, they can be used as indicators of good habitat quality, whereas some of the more eurytopic species can indicate disturbed habitats.

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1. Introduction

The ongoing growth of human populations worldwide causes a series of environmental problems, including the increasing demands for foodstuffs and natural resources, as well as the expansion of urban centers (McKinney, 2006; Lal, 2007). Urbanization causes numerous changes in the landscape through the construction of buildings, paving of streets, the reduction of native habitats, and the introduction of exotic species (Booth and Jackson, 1997). The consequences of these modifications for the species will depend on the degree of disturbance and the characteristics of the

organisms themselves. In birds, for example, species richness tends to be lower in urban areas (Cam et al., 2000). On the other hand, an apparent increase in the number of butterfly species was reported by Blair and Launer (1997) in areas with an intermediate degree of urbanization. However, the typical pattern for other invertebrates is a reduction of species richness, as seen in bees (Matteson et al., 2008), beetles (Sadler et al., 2006), and ants (Majer and Brown, 1986). Most of these studies have discussed the ability of a species to persist in urban environments as a function of its tolerance to habitat disturbance and its capacity for dispersal and to avoid urban hazards (Germaine and Wakeling, 2001).

In the specific case of aquatic environments, urban development typically leads to an increase in the quantity of sediments being deposited in rivers and streams, in addition to the reduction or elimination of the vegetation of the riparian zone. Urbanization also causes a reduction in the input of woody material, depriving these environments of debris that may help to dissipate the flow of energy and protect stream beds and margins from erosion. There is

* Corresponding author at: Laboratório de Ecologia e Conservação, Instituto de Ciências Biológicas, Universidade Federal do Pará, Rua Augusto Correia, N° 1 Bairro Guamá, CEP 66.075-110 Belém, Pará, Brazil. Tel.: +55 91 3201 8896.

E-mail addresses: csmonteirojr@gmail.com, claudiomonteiro80@hotmail.com (C.d.S. Monteiro Júnior).

also a greater input of nutrients and contaminants, leading to the eutrophication of these environments (Couceiro et al., 2007). Typically, the riparian vegetation is substituted initially by grasses with superficial roots that provide little resistance to the widening of the channel (Booth and Jackson, 1994). The canopy cover over the watercourse is also lost, eliminating the shade that regulates both temperature and the input of leaf litter into the aquatic food chain (Booth and Jackson, 1997), often resulting in greater microclimatic variation.

Many organisms are sensitive to such changes in the environment due to biological characteristics such as their naturally low abundance, microhabitat requirements or the need for relatively large areas in which to establish their populations (Henle et al., 2004). This indicates that the loss of species is not random, but rather affects organisms that share certain specific ecological traits (Williams et al., 2010) that make them more vulnerable to habitat disturbance (Berry et al., 2006). In undisturbed habitats, on the other hand, established species may be able to impede the invasion of new species, and intermediate levels of disturbance may interrupt competitive hierarchies, leading to an increase in mortality and allowing the establishment of less-competitive species (Connell, 1978). Habitat disturbance may thus alter the structure of the landscape and favor certain groups of species, depending on their habitat preferences and on their behavioral and physiological characteristics.

Communities of adult and larval odonates may be affected by modifications in environmental conditions, such as presence of riparian habitats, the hydroperiod, air temperature, the concentrations of pollutants in the water and its physical–chemical characteristics (electrical conductivity, pH, dissolved oxygen and temperature), flow rates, and discharge (Corbet, 1999; Remsburg et al., 2008; Remsburg and Turner, 2009). This sensitivity to environmental conditions and the distinct phases of the life cycle of these insects, which is characterized by aquatic immature phases and terrestrial/aerial adults, may potentially provide insights into disturbances in both aquatic and terrestrial systems (Butler and de Maynadier, 2008; Foote and Hornung, 2005; Oertli, 2008).

Odonata have a wide distribution among habitats, but specific species occur in each habitat, thus making them useful as indicators of environmental quality (Clark and Samways, 1996; Monteiro-Júnior et al., 2013; Samways and Steytler, 1996; Silva et al., 2010). Many species are habitat specialists, being sensitive to changes in the environment, and may become extinct more than generalist species (Dolný et al., 2012; Suhonen et al., 2010). In the case of damselflies (Zygoptera), species diversity appears to be influenced primarily by environmental heterogeneity (McPeck, 2008), with more complex habitats favoring the establishment of larger numbers of species, given that these insects require riparian vegetation with perches due to their reduced dispersal capacity, as well as areas for foraging and reproducing, and to provide refuge from predators (Corbet, 1999; Johansson, 2000). By contrast, the removal of riparian vegetation may be favorable to the dragonflies (Anisoptera), which are generally larger in size and more tolerant of disturbance, given that they require open areas, such as forest clearings, in which to absorb sunlight (De Marco and Resende, 2002; May, 1976, 1979, 1991). Clearing therefore leads to an increase in dragonfly species richness.

Urbanization is a major cause of environmental homogenization (McKinney, 2002), and this brings as a consequence the loss of biodiversity through decreases in the richness of many species (McKinney, 2008), including odonates (Willigalla and Fartmann, 2012). However, most studies evaluating the effects of urbanization that have been conducted in lentic environments such as lakes, ponds and temporary pools either claim that there is an increase in species richness (Ackerman and Galloway, 2003; Craves and

O'Brien, 2013) or did not find any indication of change (Goertzen and Suhling, 2013). However, information on the effect of the urbanization on species in tropical rainforest is scarce or nonexistent (Monteiro-Júnior et al., 2014).

In the present study we evaluate the effect of urbanization on adult odonates with change in the landscape. We evaluated the potential of odonate species as bioindicators of environmental quality in areas impacted by the urbanization process in the central Amazon basin. Specifically, we test the following hypotheses: (i) there is a higher proportion of sensitive species of Zygoptera than Anisoptera in pristine environments, and (ii) habitat integrity near streams is associated with higher species richness of odonates. We model environmental parameters that affect species richness, in particular Zygoptera, and identify potential bioindicator species of adult Odonata associated with the habitat integrity index in central Amazonia, for urban and pristine areas and for intermediate and degraded locations.

2. Materials and methods

2.1. Study area

The present study focused on streams in the Brazilian municipality of Manaus (Amazonas state), including modified watercourses within the urban zone and in well-preserved environments in the Ducke Forest Reserve, which is located on the outskirts of the city (Fig. 1). The climate is hot and humid equatorial, Am in the Köppen classification (Peel et al., 2007), with two seasons, a rainy season between November and June and a dry season between July and October. The streams have different levels of disturbance resulting from the discharge of domestic effluents directly into the water and from the total or partial removal of the riparian vegetation. The few well-preserved urban streams are found where the headwaters coincide with protected fragments of forest, as is the case for the campus of the Universidade Federal do Amazonas, which is free of domestic effluents (Couceiro et al., 2007).

The streams in the city of Manaus have a range of characteristics, ranging from heavily built-up areas with large quantities of refuse (considered to be degraded) to “intermediate” areas with fewer buildings and with vegetation on at least one margin, to relatively well-preserved habitats that have no domestic refuse and where there are adjacent areas of primary forest.

Brazil has 3,562,800 km² of the Amazon and state of Amazonas, with 1,531,122 km², has the largest area among Brazilian states (Skole and Tucker, 1993). Our study was done in an area of 400 km² (Fig. 1) encompassing the city of Manaus and its surroundings, this being a single sampling area with the same characteristics. The Ducke Forest Reserve, which serves as our principal site of reference, is located north of the city of Manaus. This reserve preserves its natural characteristics and represents the conditions that prevailed at the collection sites in the city before urbanization. Manaus is probably unique among major tropical cities in having a large area of protected rainforest immediately adjacent to the city, making this an ideal location for a study such as this. In 1963 when the Ducke Forest Reserve was created the city's population was estimated at about 40,000 people (Hopkins, 2005), while today it is nearly 2 million (IBGE, 2013). With this unplanned growth of population of the city of Manaus, the forest in the city has been reduced to small fragments, thus modifying these environments, including the rivers and streams that cross the city (Couceiro et al., 2007).

2.2. Data collection

The present study focused on 30 streams with differing degrees of environmental disturbance. The data were collected on the

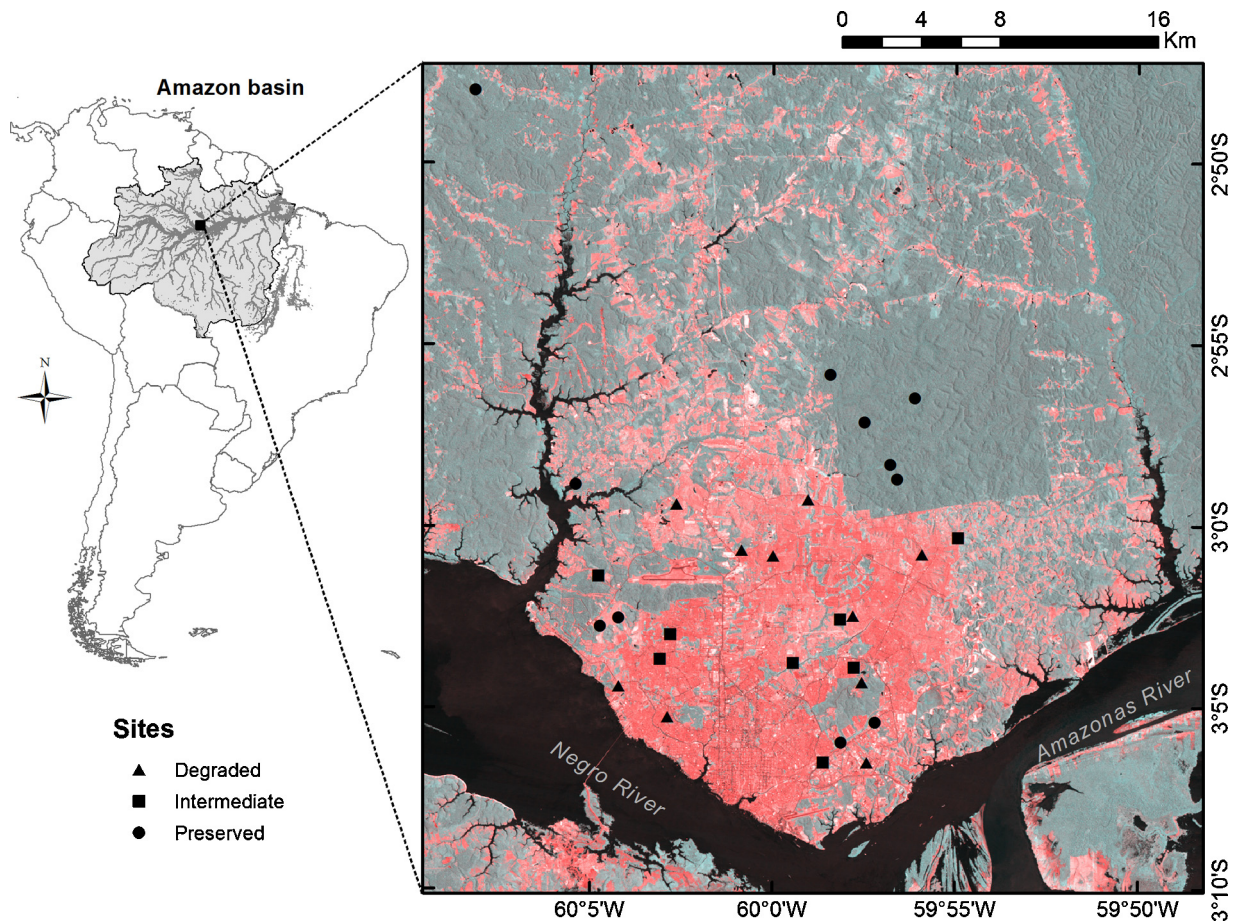


Fig. 1. Location of the urban streams in the municipality of Manaus, Amazonas (Brazil) in 2010–2011 (image: Monteiro-Júnior et al., 2014).

margins of each stream during two distinct periods representing the rainy and dry seasons, in order to provide the most complete record possible of the biodiversity at each site. Samples were collected between 10:00 h and 14:00 h only on sunny days in November and December 2010 (rainy season), and in June and July 2011 (dry season).

When prevailing conditions were inadequate (cloudy or rainy days), sampling was postponed until the subsequent sunny day. Sunny conditions are necessary to guarantee that all groups of odonates (conformers, heliotherms, and endotherms) are active during sampling period, given the thermoregulatory limitations of the adults (De Marco and Resende, 2002; May, 1976, 1991). Each site was sampled by two collectors for an average of 90 min. Adult odonates were collected following the protocol of De Marco (1998), which involves sampling a fixed area with a sweep net, over a 100-m stretch on each bank of the stream.

The captured adults were taken to the Laboratory of Cytotaxonomy and Aquatic Insects at the National Institute for Research in Amazonia (INPA), in Manaus, for identification. The specimens were stored in paper envelopes and immersed in acetone P.A. for 12 h (Zygoptera) or 24–48 h (Anisoptera) in order to maintain the coloration pattern of the specimens and guarantee their preservation (Needham et al., 2000). Species were identified using taxonomic keys (e.g. Garrison et al., 2006, 2010; Lencioni, 2005, 2006), and, whenever necessary, the specialists Frederico Lencioni (Zygoptera), Gunther Fleck (Anisoptera) and Ulisses G. Neiss were consulted. All of the specimens collected were deposited in the INPA invertebrate collection.

2.3. Habitat characteristics

The study streams have degrees of disturbance ranging from sites where the riparian vegetation had been partly or totally removed and domestic or industrial effluents are discharged into the water, to well-preserved areas located within forest fragments (see Couceiro et al., 2007). Habitat integrity was evaluated using the index developed by Monteiro-Júnior et al. (2014), which focuses on the structural characteristics of the environment on both banks of the stream (Table 1). Each of these structural variables represents a component of the index, with each component being rated on a scale of between four and six alternative scores, with the highest scores always representing the most well-preserved habitats.

For the calculation of the index, the different variables were weighted according to the scale used in each case:

$$p_i = \frac{a_o}{a_m} \quad (1)$$

where p_i = the weighted score for the i th variable, a_o = the score recorded for the variable, and a_m = the maximum possible score for the variable. The p_i values were used to calculate the HII :

$$HII = \frac{\sum_{i=1}^n p_i}{n} \quad (2)$$

where HII = the habitat integrity index, and n = the number of variables analyzed. The HII scores range from zero to one, where the highest scores represent the most well-preserved habitats (Monteiro-Júnior et al., 2014).

Table 1

Variables used for the calculation of the habitat integrity index (*HII*) of the streams surveyed in 2010/2011 in the municipality of Manaus (Amazonas, Brazil), and the contribution of each variable to the loading of the first component of the PCA.

Environmental variable	Loadings	
	Axis 1	Axis 2
Stream access (SA)	−0.897 ^a	−0.228
Width of the riparian forest (WRF)	−0.896 ^a	−0.129
Preservation of the riparian forest (PRF)	−0.871 ^a	−0.243
Condition of the riparian forest within 10 m of the stream (CRF10)	−0.906 ^a	−0.149
Water-retention mechanisms (WRM)	−0.885 ^a	−0.099
Channel structure (CS)	−0.314	0.254
Water flow in the channel (WFC)	−0.536	0.103
Canopy cover (CC)	−0.506	0.369
Absence of human occupation (AHO)	−0.897 ^a	−0.719
Absence of domestic or industrial effluents (ADIE)	−0.824 ^a	−0.276
Absence of building density (ABD)	−0.908 ^a	−0.248
Absence of dumped trash (ADG)	−0.923 ^a	−0.159
Eigenvalues	7.77	1.28
Broken-stick	3.1	2.1

^a Considering the loading above 0.7.

2.4. Data analysis

The data from the two surveys conducted at each site were combined in order to provide the most complete estimate possible of odonate diversity at each locality. Rather than an arbitrary classification of habitats, a principal components analysis (PCA), as described below, was considered to be the best approach to identify the most representative impacts of the urbanization process, with sites being classified as “preserved,” “intermediate” or “degraded” on the basis of this analysis.

Differences in the species richness of different sites based on the classification of habitat integrity (preserved, intermediate or degraded) were based on a first-order jackknife estimator of species richness with 95% confidence intervals (Colwell et al., 2004; Gotelli and Colwell, 2001), with groups being considered statistically different when there was no overlap between the confidence interval of one group and the mean score of the other.

The effects of environmental variables as pH, dissolved oxygen, air temperature, canopy, depth and width of the streams on the richness of the Odonata community were evaluated by multiple regression (Zar, 2010). To avoid multicollinearity problems, a matrix of Pearson correlations between environmental variables was calculated, when the correlation between two or more variables is equal to or greater than 0.7 only one of these was used in the analysis (Sokal and Rohlf, 1995).

The PCA was used to reduce the dimensionality of the 12 structural components of the *HII* and thus provides a more-concise visualization of the differences among the distinct types of habitat. The broken-stick criterion was used to designate the number of axes used in the ordination. In order to evaluate the differences in the environmental variables among the three habitat categories (preserved, intermediate, and degraded), the information summarized by the principal axes was evaluated using one-way analysis of variance (ANOVA) to test if the data is normally distributed and homoscedastic (Levene's test for homogeneity of variances with $p < 0.05$); otherwise a non-parametric Kruskal–Wallis test was applied (Zar, 2010). The level of significance used in all analyses was set to $p < 0.05$.

The possible existence of indicator species for each level of habitat integrity was evaluated based on Dufrene and Legendre's (1997) indicator value (IndVal), as modified by De Cáceres et al. (2010), which improves the analysis of indicator species through the combination of local groups. This index is calculated by estimating

specificity (whether the species shows an association with a certain type of habitat, occurring only in a given group or environmental condition) and fidelity (whether the species is invariably present under a given environmental condition). These two values were multiplied by 100 to provide a single score that ranges from 0 to 100, with higher values representing greater consistency in the classification of the species. Significance values were obtained from a Monte Carlo randomization test, using 10,000 replicates (Dufrene and Legendre, 1997). All of the analyses were run in the R program (R Development Core Team, 2011) using the Vegan, Stats, and Indicspecies packages.

3. Results

3.1. Odonate community

A total of 908 specimens was collected, representing 60 species, 32 genera, and 11 families. Zygoptera was the most abundant suborder, with 626 individuals belonging to 37 species, 19 genera and 8 families, and an estimated species richness of 50 ± 7.3 (mean $\pm$ confidence interval), while 282 anisopteran specimens were collected, representing 23 species, 13 genera, and 3 families, with an estimated richness of 35 ± 7.8 species.

Total odonate species richness for the study area was estimated as 85 ± 8.54 . The most abundant zygopteran species were *Argia* sp. 2 (102 individuals), *Argia* sp. 1 (84), *Hetaerina sanguinea* Selys, 1853 (66), and *H. moribunda* Hagen in Selys, 1853 (56), representing 59% of the total number of zygopteran specimens collected during the study. The most abundant anisopteran species were *Erythrodiplax basalis* (Kirby, 1897), with 115 individuals, *Perithemis mooma* Kirby, 1889 (71), and *Erythrodiplax fusca* Rambur, 1842 (35), accounting for 78% of the specimens representing this suborder.

Considering the different classes of habitat integrity, the highest species richness of adult odonates was recorded in the preserved streams (49.45 ± 8.15), with these sites having an average of 5 more species than intermediate sites, and 28 more than degraded streams (Fig. 2a). Considering each suborder, zygopterans had the highest species richness (36.31 ± 5.43) in preserved streams, with an average of 13 more species than in intermediate habitats, and 26 more than in degraded areas (Fig. 2b). In the anisopterans, by contrast, the highest estimated species richness (20.56 ± 5.03) was recorded at intermediate streams, with 7 species more than in preserved streams, on average, and 10 more than in degraded habitats (Fig. 2c).

Considering the correlation matrix for removal of correlated variables, three variables (mean depth, dissolved oxygen and canopy cover) were used for the multiple regression analysis of species richness. We evaluated the results of the regression-model environmental variables affected the species richness of Odonata ($R^2 = 0.375$; $F_{(3,26)} = 5.22$; $p < 0.005$; Table 2). The partial coefficients of the multiple regression were only significant for dissolved oxygen, which showed a positive correlation that explained 37% of the variation (slope = 0.644; $p = 0.001$). For Zygoptera, we obtained the same pattern, with the regression model being significant for environmental variables ($R^2 = 0.626$; $F_{(3,24)} = 13.39$; $p < 0.001$; Table 2). The partial coefficients of the multiple regression were significantly dissolved for oxygen, showing a positive correlation that explained 62% of the variation (slope = 0.736; $p = 0.001$). However, for Anisoptera the model did not show any correlation with species richness ($R^2 = 0.140$; $F_{(3,22)} = 1.20$; $p = 0.332$; Table 2).

The first PCA axis explained 64.8% of the variation in the abiotic data, and significant differences were found in the environmental characteristics of the three habitat classes ($F_{(2,27)} = 248.8$, $p < 0.001$). Significant differences ($p < 0.001$) were also found between each pair of classes, i.e., preserved vs. degraded, preserved vs. intermediate, and intermediate vs. degraded. The first PCA component

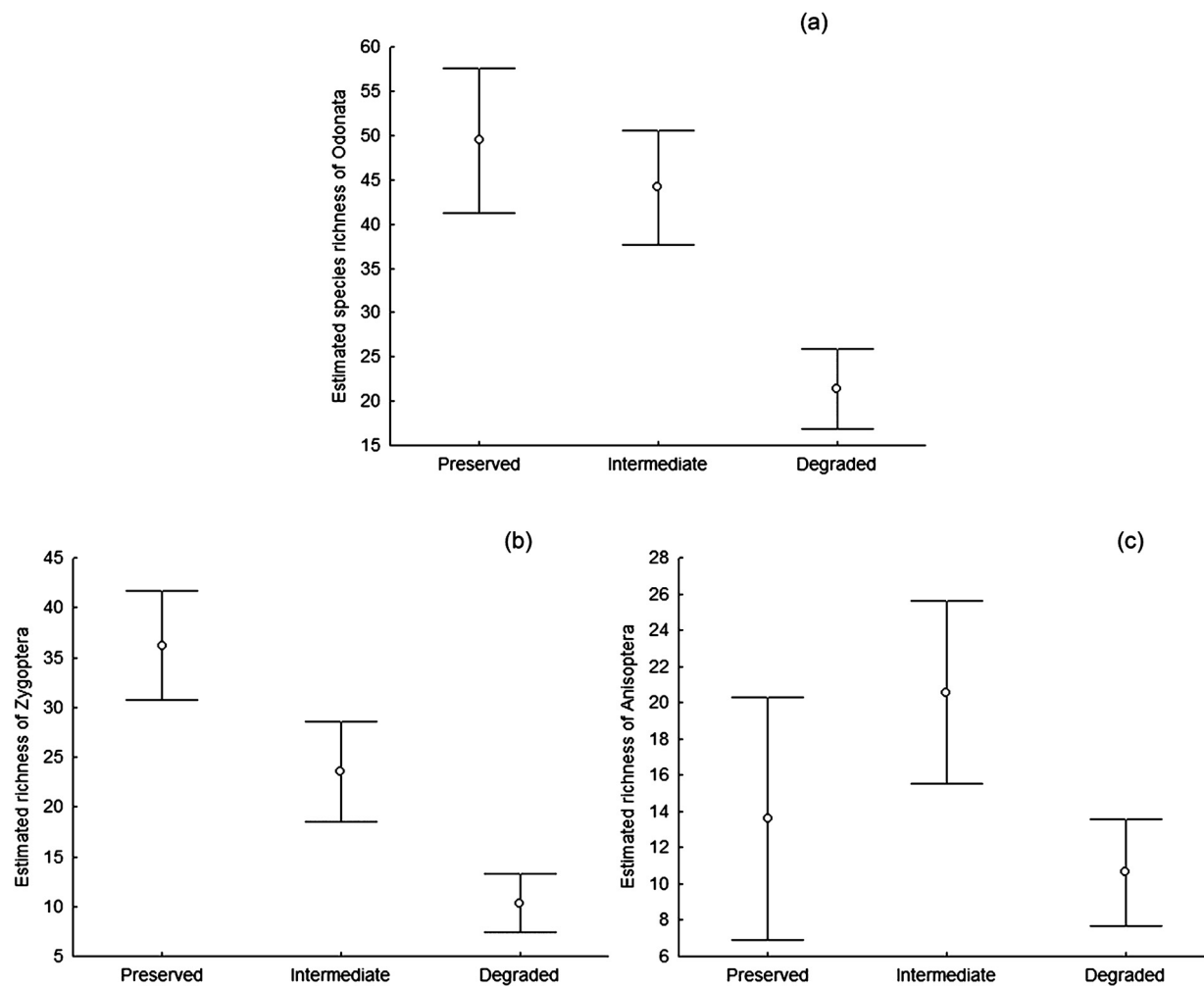


Fig. 2. Estimated species richness (first-order jackknife: mean $\pm$ confidence interval) of (a) Odonata, (b) Zygoptera, and (c) Anisoptera in the streams of the municipality of Manaus (Amazonas, Brazil) representing different degrees of habitat impact (preserved, intermediate, degraded) based on the data collected in 2010/2011.

correlated negatively with the absence of trash and buildings, the condition of the riparian vegetation within 10 m, access, width and preservation of the riparian forest, water retention mechanisms, and an absence of domestic or industrial effluents (Table 1). The preserved habitats had high scores for the correlated variables, while intermediate habitats had intermediate scores, and degraded streams had low values (Fig. 3).

Table 2
Partial coefficients of multiple regression evaluating the relationship of the total species richness of Odonata with environmental variables for streams in the city of Manaus (Amazonas, Brazil) in 2011.

	Beta	Std. Err. of Beta	t	p level
Odonata				
Mean depth (m)	0.009	0.208	0.043	0.966
O ₂	0.644	0.169	3.810	0.001
Canopy cover (%)	−0.092	0.209	−0.439	0.664
Zygoptera				
Mean depth (m)	0.088	0.154	0.572	0.573
O ₂	0.736	0.129	5.691	0.001
Canopy cover (%)	0.207	0.156	1.323	0.198
Anisoptera				
O ₂	0.196	0.209	0.939	0.358
Canopy cover (%)	−0.332	0.259	−1.282	0.213
Mean depth (m)	0.074	0.262	0.284	0.779

3.2. Distribution of species

The most widely-distributed zygopteran species recorded in the present study were *Argia bicellulata* (Calvert, 1909), found at 15 of the 30 sample points, and *Argia* sp. 1, which was recorded at 14

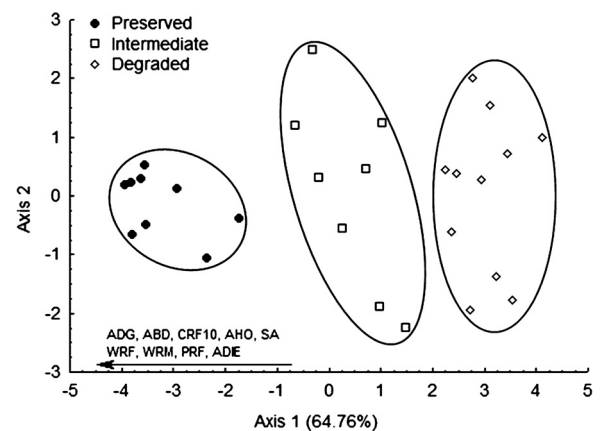


Fig. 3. Results of the principal components analysis of the *HII* variables recorded in 2010/2011 at the streams of the urban zone of the city of Manaus (Amazonas, Brazil) representing different categories of habitat impact (preserved, intermediate, degraded).

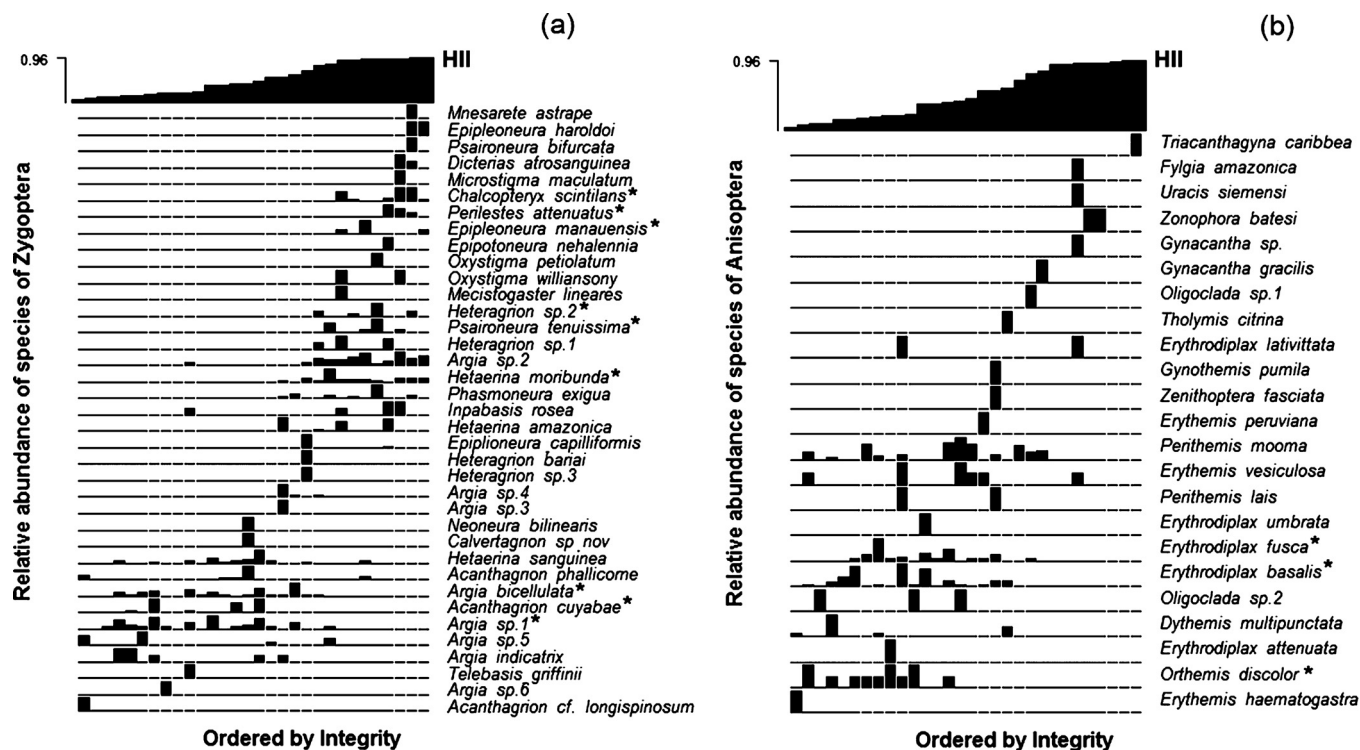


Fig. 4. Species density of (a) Zygoptera and (b) Anisoptera at the streams surveyed in the municipality of Manaus (Amazonas, Brazil) in 2010/2011 ranked by the habitat integrity index (HII). *Bioindicator species (significant IndVal).

sites. While both species were collected in all three habitat types (preserved, intermediate, and degraded), they were found much more often in intermediate or degraded streams (Fig. 4a). The most widely-distributed anisopteran was *E. basalis*, recorded at 13 sites, followed by *E. fusca* and *P. mooma*, with 12 sites; in all cases, these areas had intermediate or degraded habitats (Fig. 4b).

The majority of the species were recorded at only one site. *Epipotoneura nehalennia* Williamson, 1915, *Mnesarete astrape* De Marmels, 1989, *Psaironeura bifurcata* (Sjöstedt, 1918), *Heteragrion bari* De Marmels, 1989, and *Oxystigma petiolatum* (Selys, 1862) were all recorded once in association with preserved streams. *Telebasis griffinii* (Martin, 1896), *Acanthagrion cf. longispinosum*, *Neoneura bilinearis* Selys, 1860, and *Calvertagrion sp. nov.* were all found only in intermediate or degraded habitats.

Microstigma maculatum Schmidt, 1958, and *Mecistogaster lineares* (Fabricius, 1776) are typically found in association with phytotelmata, and are thus dependent on forested environments. These species were recorded only in the two Acará microbasins in the Ducke Reserve, these being streams that are protected by a dense and well-preserved Amazonian forest.

3.3. Indicators of environmental quality

Overall, nine (15%) of the 60 species analyzed demonstrated potential for use as bioindicators of environmental quality, and four were characteristic of disturbed or degraded habitats. Seven species (*Perilestes attenuatus* Selys, 1886, *Epipleoneura manauensis* Santos, 1964, *Heteragrion sp. 2*, *Psaironeura tenuissima* [Selys, 1886], *Chalcopteryx scintillans* Selys, 1886, and *H. moribunda*, *Argia sp. 2*) were indicators of preserved habitats, two (*Orthemis discolor* [Burmeister, 1839], *Acanthagrion cuyabae* Calvert, 1909) of disturbed habitats, and four (*A. bicellulata*, *E. basalis*, *Argia sp. 1*, *E. fusca* Rambur, 1842) of degraded habitats (Table 3).

The highest-ranking species in terms of environmental specificity and site fidelity were *Argia sp. 2*, which scored 96.1%

Table 3

Indicator values (IndVal) for the odonate species collected at the streams surveyed in 2010/2011 in the municipality of Manaus (Amazonas, Brazil).

Species	IndVal	Ranking	p < 0.05	Habitat association
<i>Orthemis discolor</i>	0.715	8	0.012	Intermediate
<i>Acanthagrion cuyabae</i>	0.593	11	0.041	Intermediate
<i>Argia sp. 2</i>	0.961	1	0.001	Preserved
<i>Hetaerina moribunda</i>	0.881	2	0.001	Preserved
<i>Chalcopteryx scintillans</i>	0.775	6	0.002	Preserved
<i>Psaironeura tenuissima</i>	0.632	9	0.011	Preserved
<i>Heteragrion sp. 2</i>	0.632	10	0.009	Preserved
<i>Epipleoneura manauensis</i>	0.548	12	0.042	Preserved
<i>Perilestes attenuatus</i>	0.548	13	0.047	Preserved
<i>Argia bicellulata</i>	0.827	3	0.002	Degraded
<i>Erythrodiplax basalis</i>	0.806	4	0.005	+ Intermediate Degraded
<i>Argia sp. 1</i>	0.797	5	0.003	+ Intermediate Degraded
<i>Erythrodiplax fusca</i>	0.731	7	0.039	+ Intermediate Degraded

(preserved streams), *H. moribunda*, with 88.1% (preserved), *A. bicellulata*, with 82.7% (intermediate and degraded), and *E. basalis*, with 80.6% (intermediate and degraded) (Table 3).

4. Discussion

4.1. Effects of urbanization on the species richness of adult odonates

A large number of studies have described the effects of urbanization on species richness in both plants and animals, and have shown that the process may result in either an increase or a decline in richness, depending on its intensity and the taxonomic group analyzed (Chace and Walsh, 2006; Dickman, 1987; McKinney, 2008). The present study recorded a decrease in odonate species richness

in areas that are more intensely impacted by the urbanization process (degraded streams), a pattern expected for the majority of organisms (McKinney, 2008).

Contrasting results were recorded for the two odonate suborders analyzed in the present study. Whereas the number of zygopteran species declined gradually with urbanization, as observed in other groups of animals, such as birds (Chace and Walsh, 2006), mammals, reptiles, and amphibians (Dickman, 1987), anisopterian species richness increased in intermediate habitats, as predicted by Connell (1978). When the disturbance was more intense, however, even anisopterian richness decreased, with this group being reduced to a few highly tolerant species, resulting in a high degree of homogenization of the community. An apparent increase in species richness under intermediate levels of habitat disturbance has been recorded in a number of organisms, including mammals (Racey and Euler, 1982), reptiles (Germaine and Wakeling, 2001), and butterflies (Blair and Launer, 1997). Alterations in the environment, such as an increase in the productivity of the water and the substitution of native vegetation by exotic species lead to an increase in temperatures and increasing exposure of the soil to sunlight, which may favor the anisopterians in particular, given their need to thermoregulate (May, 1979, 1991).

The anthropogenic factors typical of the urbanization process were absent in preserved habitats, which were in primary forest. Stenotopic species were found only in these habitats. By contrast, sites with riparian vegetation, even narrow strips of habitat with human impact, as seen in the intermediate habitats surveyed in the present study, have a typical fauna that is characterized primarily by zygopteran species. It is thus clear that the reduction in species richness and alterations in community composition are caused in particular by the reduction or elimination of the vegetation, especially the habitats associated with rivers and streams (Couceiro et al., 2007). In other words, the integrity of these aquatic habitats depends fundamentally on the preservation of the adjacent vegetation (Allan and Johnson, 1997; Dudgeon et al., 2006). This is not only due to the role of this vegetation as a buffer zone, but also because it provides an alternative source of energy for the watercourse through the input of leaves, branches, and fruits.

Dissolved oxygen was positively correlated with species richness of Odonata, especially for Zygoptera, which is the more sensitive of the two suborders to changes in the environment (Corbet, 2006; Butler and de Maynadier, 2008). A previous study in the same area has demonstrated that low dissolved oxygen concentrations decrease species richness of macroinvertebrates and that sites with high oxygen concentrations were positively correlated with high species richness (Couceiro et al., 2007). This is because the low concentration of dissolved oxygen is directly linked to high temperatures due to deforestation (Allan and Castillo, 2007). Many odonates cannot survive this combination, especially species of Zygoptera.

4.2. Bioindicator species

Biological communities tend to be adapted to specific ecological conditions and will present varying levels of tolerance to modifications in the environment (Alba-Tercedor, 1996; Wiens et al., 2010). In the present study, for example, some species, such as *O. discolor*, were indicators of an intermediate degree of urbanization. This genus appears to be typical of standing water habitats, and is made up of large-bodied species that spend most of their time flying, and can be found in calm stretches of watercourses, lakes, and pools (De Marmels, 1988; Pinto, 2011). The changes resulting from deforestation and urbanization lead to an increase in temperatures and the frequent reduction of stream flow in many areas, which may favor the species in this genus. Many of the streams in intermediate habitats still had some vegetation along their banks,

which may have contributed to the establishment of zygopteran species such as *A. cuyabae*. This reflects the importance of this type of vegetation, even when reduced in extent, for the movement of species between areas, and for providing adequate conditions for their ecophysiological requirements (Stewart and Samways, 1998).

E. fusca, *E. basalis*, *A. bicellulata* and *Argia* sp. 1 are indicator species of disturbed habitats. These species had the greatest degrees of fidelity and specificity to this type of habitat, indicating their ability to tolerate the impacts inherent in the urbanization process. Species of the genus *Erythrodiplax* appear to be typical of lentic systems (Ferreira-Peruquetti and De Marco, 2002) and may thus have taken advantage of the changes provoked by the urbanization process to occupy these environments, although even this genus showed reduced species richness in the most intensely urbanized areas. Previous studies in the same region confirmed that *E. basalis* is an indicator of disturbed habitats (Monteiro-Júnior et al., 2013). The colonization of lentic systems by this species within lotic environments may be accounted for by the changes in the environment (Ferreira-Peruquetti and De Marco, 2002). These changes include the silting caused by deforestation, which ultimately results in the transformation of these environments into semi-lotic systems due to the reduction in stream flow and the formation of large areas of still water. The wider distribution of these species may be related to their more ample niches. As most of the sites at which *A. bicellulata*, *Argia* sp. 1, *E. basalis*, and *E. fusca* were found were either degraded or intermediate, they may be considered to be generalist species (Andrew and Hughes, 2005; Wilson and Yoshimura, 1994). Furthermore, it is noteworthy that the population dynamics of landscape-scale in humans impact upon the species in which the movement of individuals among populations may avoid local extinction (Lowe, 2002). For example, in a study performed in France, the species *Calopteryx exulans* Harris, 1782 was able to disperse between and along watercourses, with overland dispersion occurring also, although no environmental impact was reported (Chaput-Bardy et al., 2008). Thus, the small riparian vegetation found in our study (some intermediates and degraded) helped in maintaining the dispersion of some species, as mentioned above, since the removal of riparian vegetation may isolate and extinguish species populations (Lowe, 2002).

Where the urbanization process is accentuated, as in the degraded sites in the present study, even the most tolerant species become rare or locally extinct. A similar pattern has been recorded in studies of other organisms, such as birds (Marzluff, 2001), mammals (Mahan and O'Connell, 2005), reptiles (Germaine and Wakeling, 2001), and insects (Blair and Launer, 1997), including odonates (Willigalla and Fartmann, 2012). However, some studies in ponds and wetlands in Europe and North America found an increase or no change in species richness of odonates with increasing urbanization (Craves and O'Brien, 2013; Goertzen and Suhling, 2013), but these species are less susceptible to changes in the environment than are forest species, since species characteristic of ponds and temporary puddles can live at high temperatures and with low concentrations of oxygen available for their offspring. Moreover, odonates in lentic environments are adapted to survive in temporary habitats and so have very fast larval development, tolerate large environmental variations and have a great capacity for dispersal that allows migration to areas with more favorable environmental characteristics (Corbet, 1999).

Species that have restricted geographic ranges are often vulnerable to habitat disturbance. Given this, the relative proportion of endemic to generalist species may be a useful parameter for the evaluation of the conservation status of an area (Simaika and Samways, 2009). In the present study, all of the indicator species of preserved habitats (*P. attenuatus*, *E. manauensis*, *Heteragrion* sp. 2, *P. tenuissima*, *C. scintillans*, *H. moribunda*, and *Argia* sp. 2) belong to the suborder Zygoptera. Most zygopterans are specialists and lay their

eggs within the vegetation (endophytic). They also require vegetation on which to perch, spot their prey, and thermoregulate (Corbet, 1999; Samways and Sharratt, 2010). They are thus more vulnerable to changes in the vegetation and may often be excluded from disturbed habitats or even become locally extinct. A species restricted to a given habitat type may represent a better ecological indicator of environmental disturbance than a habitat generalist, due to its greater susceptibility to local or regional extinction (Carignan and Villard, 2002; De Cáceres and Legendre, 2009).

The community of adult odonates is a potentially useful indicator of environmental quality, given that different species may be restricted to specific sites or groups of sites. Modifications in the species composition of the community may thus indicate specific types of disturbance in the biological characteristics of a stream or in the surrounding area, due to the removal of the vegetation, changes in land use, and pollution of the water, in a way that variables representing the physical characteristics of the environment might not identify (Clark and Samways, 1996; Stewart and Samways, 1998; Butler and de Maynadier, 2008).

The results of the present study provide important guidelines for the classification of habitats and development of conservation strategies, given that the associations observed in the odonates may also be typical of other invertebrate groups, or even of other types of organisms. While urbanization generally leads to a loss of biodiversity, the present study showed that one of the two odonate suborders, the Anisoptera, may become more diverse under intermediate levels of urbanization. By contrast, only zygopterans showed a high degree of fidelity and specificity to preserved habitats, although a small number of zygopteran species had this same type of relationship with intermediate or degraded habitats. Anisopterans only showed high fidelity and specificity to intermediate and degraded habitats. The results of this study showed a higher proportion of Zygoptera in pristine environments. However, different zygopteran species may show an affinity with either preserved or disturbed habitats, reflecting the ecological diversity of the species in this suborder. Among anisopterans, the indicator species were only associated with disturbed habitats. Thus, the study confirms the potential of analyzing the community of adult odonates as a rapid and effective tool for habitat assessment. The more stenotopic species can be used as indicators of good quality habitats, while the more eurytopic species may indicate disturbed environments.

Overall, the study has shown that analysis of the odonate community combined with procedures for the evaluation of environmental quality, such as the *HII*, can provide valuable guidelines for the evaluation of patterns of species richness. The composition of the community found in a given type of habitat is an efficient indicator of potential changes in the environment on a local scale. However, urban planning is necessary to avoid unregulated population growth and urbanization, and thus limit the degradation of the environment. Degradation of aquatic habitats may be impeded by the maintenance of riparian vegetation. Improvement of the environmental quality of degraded areas may prevent species substitutions and local extinctions.

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